Unmasking the State Effect: Explaining Why COVID-19 Mask Compliance Differed Across Australian States

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1 Declaration

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2 Abstract

Mask-wearing compliance during the COVID-19 pandemic varied substantially across Australian states, and Ryan et al. (2025) found that state of residence was among the most important predictors of compliance in their machine learning models, yet excluded it from their main results because a geographic label explains nothing about the mechanism behind it. This report tests whether that state effect can instead be explained by measurable, time-varying policy indicators (from the Oxford COVID-19 Government Response Tracker) and demographic indicators (from the 2021 Australian Census), using the same national survey data and three modelling approaches: logistic regression, a classification tree, and a random forest. Replacing state with these measurable factors matched the predictive performance achieved when state was included directly (random forest AUC 0.911 versus 0.906), and face-covering policy stringency emerged as the dominant replacement predictor, closely tracking real mandate and outbreak events such as Victoria's first mask mandate in 2020, the Delta-driven lockdowns of mid-2021, and the reinstatement of masks in December 2021 once the Omicron variant emerged. These results indicate that the state effect identified by Ryan et al. (2025) is substantially attributable to differences in state government policy rather than an unexplainable geographic factor, although a single classification tree consistently underperformed the other two models, particularly once state was replaced by continuous policy measures it could not split on as effectively.

3 Introduction

On 23 July 2020, Victoria became the first Australian state to make mask-wearing compulsory in public, during a second wave of COVID-19 centered on Melbourne (Costantino and Raina MacIntyre 2021, 1). Seventeen months later, in December 2021, New South Wales and Victoria reinstated mask mandates that had just been relaxed weeks earlier, this time in response to the Omicron variant, even though more than 90% of eligible adults had by then received two vaccine doses (Gulf News 2021). Between those two mandates, Australian states differed sharply, and repeatedly, in how strictly mask-wearing was required and how consistently people actually complied. Understanding why is the subject of this report.

Mask-wearing was one of the most visible and studied health behaviours of the pandemic. Research has linked compliance to demographic factors such as age and gender (Haischer et al. 2020), attitudinal factors such as perceived personal risk (Binter, Pešout, et al. 2023; Schumpe et al. 2022), and the strength of government policy (Wismans et al. 2022). Most of this research is about individuals or about whole countries; little asks whether variation *within* one country, between its states, can itself be explained by measurable factors rather than treated as an unexplained label.

Ryan et al. (2025) raised exactly this question using the same Australian survey data used here. Modelling mask-wearing compliance with four machine learning algorithms, they found state of residence to be one of the most consistently important predictors across every model, often more important than age, gender, or perceived risk. Because a state is simply a label, this does not explain the mechanism behind it. The authors excluded state from their reported results and called for further work to disentangle it.

This report answers that call directly, described fully in Methods. A No State model shows what individual-level factors alone can achieve. A model including state as a predictor reproduces Ryan et al. (2025)'s approach. A third model then removes state and replaces it with measurable, time-varying government policy indicators and 2021 Census demographic indicators, testing whether they explain what state used to represent. Three algorithms, logistic regression, a classification tree, and a random forest, are fitted throughout, so the state effect and its explanation can be checked across different modelling assumptions rather than depending on one algorithm.

Background reviews the relevant literature and the three modelling approaches. Methods describes the data and models in enough detail to reproduce the analysis. Results reports model performance and predictor importance. Discussion interprets these results, including the real historical events that plausibly explain the patterns found.

4 Background

4.1 Mask-Wearing Compliance During COVID-19

A consistent body of research identifies demographic, attitudinal, and policy-related predictors of mask-wearing compliance, though findings vary with how compliance was measured and where each study was conducted.

Haischer et al. (2020) Results directly counted mask use among 9,935 shoppers entering US retail stores between June and August 2020, avoiding self-report bias. Women were about 1.5 times more likely than men to be wearing a mask, older shoppers had higher odds of compliance than younger ones, and urban/suburban shoppers were roughly four times more likely than rural shoppers to be masked. Once local mandates were introduced, compliance rose above 90% across all demographic groups. This is an early demonstration that policy can compress gaps between demographic groups that otherwise predict compliance strongly, a pattern this project investigates at the Australian state level.

Cunningham and Nite (2021) Results took a complementary approach, modelling compliance at the US county level using a two-level random-effects regression. County-level health-behaviour indices were positively associated with mask-wearing, while the physical environment held a negative association. This is one of few studies to model compliance as a property of *place* as well as of individuals, directly analogous to the state-level component of this project.

At the individual level, Binter, Pešout, et al. (2023) and Schumpe et al. (2022) point to psychological rather than purely demographic drivers. Binter, Pešout, et al. (2023) Results identified risk perception and general health-behaviour orientation as significant motivators of mask-wearing. Schumpe et al. (2022) Results followed longitudinal international survey data ($N = 3{,}040$) across multiple waves and found predictors of adherence remained largely stable over time, rather than shifting as the pandemic progressed.

The role of government policy specifically has also been documented directly. Wismans et al. (2022) Results studied approximately 7,000 university students across multiple countries and found country-level policy stringency strongly and positively associated with face mask use. Critically, the strength of the relationship between an individual's own risk perception and their behaviour was itself moderated by their country's policy stringency: in countries with strict mandates, individual attitudes mattered less than in countries without regulation. This finding, that policy context changes how much individual attitudes matter rather than simply adding to them, is the mechanism this project investigates at the Australian state level.

This project sits at the intersection of two ideas drawn from the literature above. Following the place-based approach of Cunningham and Nite (2021), it treats jurisdiction of residence as a place-level effect worth explaining in its own right, using measurable place-level factors rather than a fixed grouping label. Following Wismans et al. (2022)'s finding that policy context shapes how much individual attitudes matter, it combines individual-level survey data with objective, time-varying government policy-tracking data, extending that approach to the more granular within-country level of Australian states rather than whole countries. Using Australian states as the unit of geographic variation and government policy-tracking data as the measurable mechanism, this project directly tests whether the state-level differences documented by Ryan et al. (2025) can be explained this way.

4.2 The Ryan et al. (2025) Study

Ryan et al. (2025) analysed the same Australian COVID-19 mask compliance dataset used in this project, the YouGov Behaviour Tracker (described in Section 5.1), asking how predictors of mask-wearing and general protective-behaviour adherence differed before versus after mandates were introduced. They compared four algorithms, logistic regression, a classification tree, random forest, and XGBoost, via cross-validation for two outcomes across two time periods. Random forest and XGBoost consistently outperformed the other two and were carried forward to a final stage of eight fitted models, from which stable feature-importance rankings were extracted by refitting each model 10,000 times with random up-sampling of the minority class.

Their central finding was that state of residence was highly influential throughout (Ryan et al. 2025, 8–9). The eight variables representing Australia's states and territories comprised between one and six of the ten most consistently important predictive features across the eight final models, and every state or territory except the Northern Territory appeared at least once in a top-ten list. Because a categorical geographic label conveys no information about *why* one state differs from another, the authors excluded it from their main reported results and named disentangling the state effect as an important direction for future research (Ryan et al. 2025, 13).

Beyond the state effect, Ryan et al. (2025, 1) found that general protective-behaviour compliance, survey week, age, non-household contacts, wellbeing, and perceived illness threat were common predictors across both time periods, while trust in government and employment status mattered before mandates only, and willingness to self-isolate became important only after mandates were introduced. This suggests that once a behaviour becomes legally required, the attitudes that previously explained voluntary compliance are partly displaced by attitudes specific to the mandate context.

This project is a direct, focused follow-up to the open question Ryan et al. (2025) leave unanswered. Rather than splitting the analysis by mandate period as they did, this report examines what state is allowed to represent in the model. State is first included as an unexplained label, reproducing the finding of Ryan et al. (2025), and is then removed and replaced by a set of measurable policy and demographic quantities, directly testing whether the state effect is more than a label.

4.3 Logistic Regression

Logistic regression models the probability π that a binary outcome equals 1 (here, mask compliance) as a function of predictors $x_1, \ldots, x_p$ (formulation adapted from Data Taming course materials, (University of Adelaide, School of Mathematical Sciences 2024a, 15–16)). Rather than modelling π directly, it models the log-odds of π, known as the logit function, as a linear combination of the predictors:

$$\text{logit}(\pi) = \log\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p. \tag{1}$$

Rearranging Equation 1 for π gives the fitted probability directly, a sigmoid (logistic) curve bounded between 0 and 1:

$$\hat{\pi} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}}. \tag{2}$$

Coefficients $\beta_0, \ldots, \beta_p$ are estimated by maximum likelihood. A positive β_j indicates that increasing x_j increases the odds of mask compliance, holding the other predictors fixed; the reverse holds for a negative β_j. Predicted probabilities from Equation 2 are converted to class predictions by thresholding at 0.5.

4.4 Classification Tree

A classification tree predicts a categorical outcome by repeatedly splitting respondents into groups using a series of either/or questions (node and splitting terminology adapted from Data Taming course materials, (University of Adelaide, School of Mathematical Sciences 2024b, 11–12)). The tree begins with a single root node containing all training respondents. At each step, it considers every possible either/or question it could ask of every predictor, and chooses whichever question makes the two resulting groups as internally alike as possible in terms of the outcome.

To decide how good a candidate split is, the tree measures how "mixed" each resulting group is using a simple score called Gini impurity. A group scores low on this measure when almost everyone in it gave the same answer, and scores high when the group is an even mix of compliant and non-compliant respondents. At every step, the tree picks whichever split leaves the two new groups as unmixed as possible, and repeats this process again on each new group until a stopping rule is reached. A group that is not split any further becomes a terminal node, predicting whichever outcome was most common among the respondents who ended up there. A new respondent is classified by following the same sequence of questions from the start of the tree down to a terminal node.

Figure 1 shows an example of a fitted classification tree from this project. The root node at the top asks the first, most useful either/or question. Each subsequent box down the tree asks a further question, splitting the group again, and the coloured boxes at the bottom are terminal nodes, each showing the predicted outcome and the number of training respondents who ended up there.

Unlike logistic regression, a classification tree makes no assumption that predictors combine in a straight-line way. It also picks up on interactions between predictors automatically, simply by asking about one predictor after another along the same branch. Its main weakness is a tendency to overfit, carving out rules that match the training data closely but do not generalise well to new respondents, which motivates the random forest below.

4.5 Random Forest

Random forest (Breiman 2001, 5) fixes the single tree's overfitting problem by growing hundreds of trees instead of one, and having them vote together rather than trusting any single tree's answer. Two sources of randomness make this work.

The first is that each tree is trained on its own random sample of respondents, drawn from the training data so that some respondents are picked more than once and others are left out entirely.

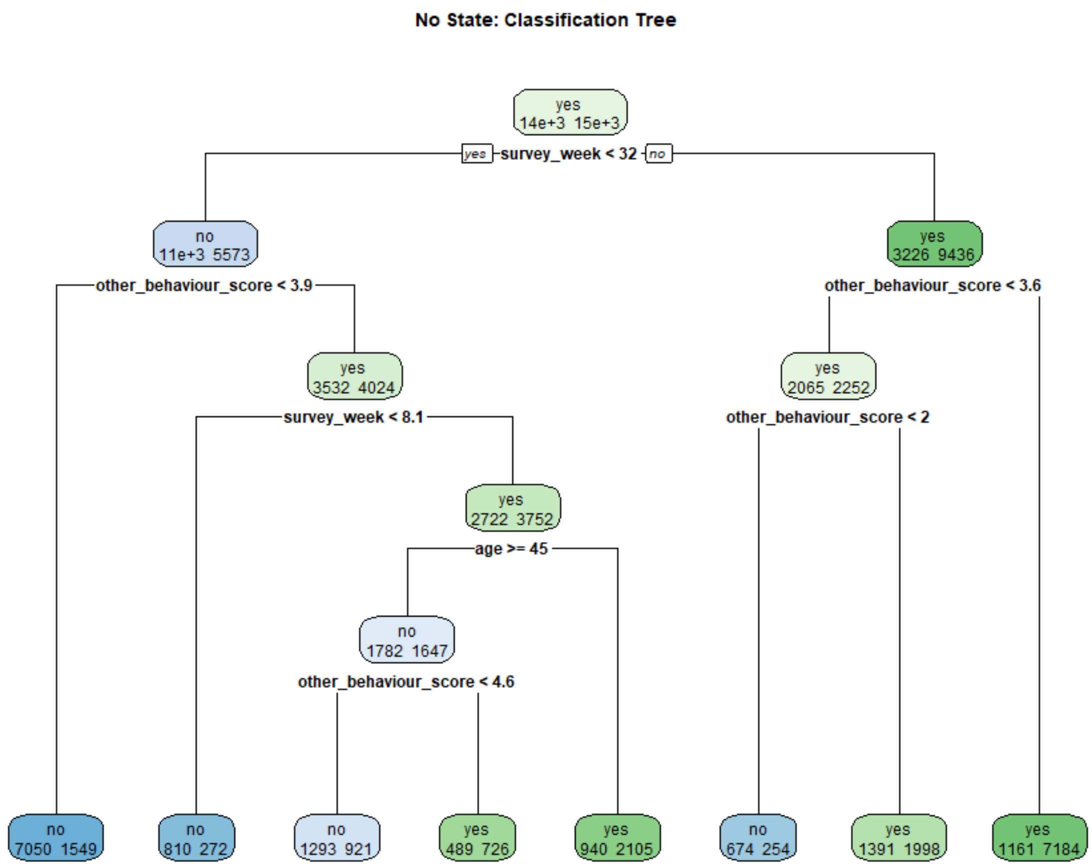

Figure 1: An example classification tree fitted in this project, using individual-level predictors only. Each box shows the split question, the respondent counts reaching that point, and the predicted class. Reading from the root node down to any terminal node gives the sequence of either/or questions used to classify a respondent.

This means every tree learns from a slightly different version of the data, so the trees end up different from one another rather than all making the same mistakes.

The second is that at each split, a tree is only allowed to choose from a small random subset of predictors, rather than all of them. This stops one or two strong predictors from dominating every tree's first split, which would otherwise make the trees look alike despite the random resampling.

A prediction for a new respondent is made by asking every tree to classify it and taking the majority vote. Because any one tree's mistakes are unlikely to be shared by most other trees, averaging across many trees cancels out much of that noise while keeping the genuine, consistent patterns.

Predictor importance is measured in a similarly simple way. One predictor's values are randomly shuffled among respondents, breaking any real relationship it had with the outcome, and the resulting drop in prediction accuracy is recorded. A large drop means the predictor was doing real work. Little change means it was not contributing much.

4.6 Evaluation Metrics

Definitions

Let TP, FP, TN, FN denote true positives, false positives, true negatives, and false negatives on the test set. The four metrics used throughout this report are:

- **AUC** (area under the ROC curve) — the probability that the model assigns a higher predicted compliance probability to a randomly chosen compliant respondent than to a randomly chosen non-compliant one. Ranges from 0.5 (random guessing) to 1.0 (perfect). Primary comparison metric, following Ryan et al. (2025). As a rough guide, AUC = 0.5 indicates no discrimination, (0.5, 0.7) poor discrimination, (0.7, 0.8) acceptable discrimination, (0.8, 0.9) excellent discrimination, and above 0.9 outstanding discrimination (interpretation scale adapted from Data Taming course materials, (University of Adelaide, School of Mathematical Sciences 2024a, 32)).

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \quad \text{Accuracy} = \frac{\text{TP} + \text{TN}}{N} \tag{3}$$

- **Sensitivity** — of all respondents who actually wore masks, the proportion correctly identified as compliant.

- **Specificity** — of all respondents who did not wear masks, the proportion correctly identified as non-compliant.

- **Accuracy** — of all respondents, the proportion whose mask-compliance status, compliant or not, was correctly predicted.

Full definitions of every variable used in this project, including exact scales and roles, are given in Appendix E.

5 Methods

5.1 Data Sources

Survey data. The primary data source is the YouGov COVID-19 Behaviour Tracker for Australia (Jones, Imperial College London Big Data Analytical Unit, and YouGov Plc 2020), comprising 53,833 survey responses collected approximately fortnightly from 1 April 2020 to March 2022, across all eight Australian states and territories. This is the same dataset analysed by Ryan et al. (2025).

Policy data. State-level policy stringency was obtained from the Oxford COVID-19 Government Response Tracker (Hale et al. 2021, 530; Oxford COVID-19 Government Response Tracker (OxCGRT) 2021): a face-covering policy score, an ordinal scale from 0 (no policy) to 4 (required with enforcement), and a stay-at-home requirement score, an ordinal scale from 0 (no restrictions) to 3 (required, with minimal exceptions), both recorded daily for each state. Only state-level totals were used; sub-state regional and city-level rows were excluded.

Demographic data. Three state-level demographic indicators, median weekly household income, percentage with a Bachelor's degree or higher, and percentage born overseas, were derived directly from the Australian Bureau of Statistics 2021 Census DataPacks (Australian Bureau of Statistics 2021), at the state/territory level, rather than from a secondary summary source. This is not an official ABS product. The demographic indicators table used in this project was compiled by the authors from the raw DataPack tables and checked against them before use, with the full derivation and a correction record given in Appendix D.

All data loading and cleaning steps are documented in full, with code, in Appendix A.

5.2 Data Cleaning

Blank strings. Several columns encoded missing values as a single blank space rather than a true empty value. All character columns were checked for this and converted to a proper missing-value code where found.

Type conversion. Numeric variables that had been stored as text were converted to numbers. The perceived severity variable required an additional text-parsing step to extract the numeric rating embedded in labels such as "5 - Moderate".

Date and survey week. The date portion of each response's time stamp was extracted and converted into a proper date. Survey week was then defined as the number of complete two-week periods that had elapsed since 1 April 2020:

$$\text{survey week}_i = \left\lfloor \frac{\text{survey date}_i - 1 \text{ April } 2020}{14 \text{ days}} \right\rfloor . \tag{4}$$

Likert re-coding. All 13 other-protective-behaviour questions (a five-point frequency scale) were re-coded to the integers 1 to 5. The four mental-health questions (a four-point scale) were re-coded to the integers 0 to 3.

Household size and employment. The response “8 or more” was re-coded to 8; “Prefer not to say” and “Don’t know” were treated as missing. Employment status was collapsed from many free-text categories into five groups by matching key phrases in each response.

Outlier removal. Values above 10 on the trust-in-government and perceived-susceptibility scales (both intended to run 0-10) were treated as missing, since values above 10 indicate a non-response code rather than a genuine rating.

5.3 PHQ-4 Structural Missingness

The mental health module was not administered to all respondents. Rather than applying a complete-case rule that would discard 23% of the sample, each respondent was assigned a three-level status: *Answered*, *Declined*, or *Not available*. This mental health response status is used directly as a predictor, in place of the underlying scores. The exact count in each category is given in Appendix A.

5.4 Variable Exclusions

Four variables were excluded from modelling due to high missingness: trust in government (42.6%), ease of self-isolation (35.1%), perceived susceptibility (33.3%), and non-household contacts (33.3%). The latter two were missing for exactly the same 17,901 respondents, indicating a shared survey-routing decision. No imputation was performed.

5.5 Outcome Variable

Mask compliance was measured using four survey items: mask use outside the home, in a supermarket, in a clothing shop, and on public transport. These are the correct items used by Ryan et al. (2025), correcting an error in the original Part A submission, which had used an earlier, incorrect set of items. Each was re-coded 1-5 and averaged; respondents scoring ≥ 4 were classified as compliant. Three of the four items were not part of the survey until wave 6 (approximately late June 2020), so the effective analysis window begins there.

5.6 Final Analytical Sample

A complete-case filter requiring all modelling variables to be non-missing yielded **41,168 respondents** from an initial 53,833 (76.5% retained). See Appendix A for the code and output that produced this figure.

5.7 Model Datasets

Three model datasets were constructed, and three models, logistic regression, a classification tree, and a random forest, were fitted to each.

- **No State** (10 predictors): survey week, age, gender, household size, employment status, wellbeing, perceived severity, willingness to self-isolate, mental health response status, and behaviour score. State of residence excluded entirely. Establishes what individual-level factors alone can explain.

- **With State** (11 predictors): the same 10 predictors plus state. Reproduces the approach of Ryan et al. (2025) and establishes the performance ceiling.

- **Measurable Factors** (14 predictors): state replaced by mask-policy strictness, lockdown strictness, median weekly household income, percentage with a Bachelor's degree or higher, and percentage born overseas. Tests whether the state effect is explainable by measurable factors.

5.8 OxCGRT Policy Variable Construction

A two-week period index was computed for both the daily government policy series and each respondent's survey date, using the same definition as survey week above (Equation (4)). Within each state and two-week period, the daily values of the two policy indicators (face-covering strictness and stay-at-home strictness) were averaged, giving the state-and-fortnight-level policy scores used in the measurable-factors model. These averaged policy scores were then merged onto each respondent's row by matching on their state and survey period (this merge step is not covered by the course's standard function reminder sheet, and was necessary because no reminder-sheet function can perform a key-based merge between two datasets with different numbers of rows). This merge was validated to confirm no respondents were lost and no missing values were introduced. The validation code and its output are given in full in Appendix C.

5.9 Train/Test Split and Model Parameters

Each of the three model datasets received its own independent 70/30 split, with the same random seed (1914220) set immediately before each split. Because each split is generated independently, the specific respondents held out for testing differ slightly across phases; AUC comparisons in Table 1 should therefore be read as comparisons of overall predictive performance rather than performance on one identical held-out set.

5.10 Model Implementation and Justification

Three models spanning different assumption structures were fitted to each dataset: logistic regression (linear in the log-odds, Equation 1), a classification tree (nonparametric, no linearity assumption, but a single greedy sequence of splits), and a random forest (nonparametric, an ensemble average of many trees). Fitting all three to every phase, rather than a single "best" algorithm, allows the state effect and its explanation by measurable predictors to be checked for consistency across genuinely different modelling assumptions rather than being an artifact of one algorithm's particular biases.

All three models were fitted using standard default settings for each algorithm; no hyperparameter tuning was performed. This is a limitation, since Ryan et al. (2025, 7–8) tuned each algorithm's settings and reported that this materially affected the relative ranking of random forest and XGBoost. Default settings were used identically across all three phases so that any AUC differences between phases reflect the predictor set available rather than differential tuning effort, isolating the comparison of interest (does replacing state with measurable predictors change performance?) at the cost of not establishing the best achievable performance for any single model.

Logistic regression's core assumption, that predictors combine linearly and additive on the log-odds scale (Equation 1), was not formally tested. If this assumption is violated for predictors such as survey week or mask-policy strictness, logistic regression would be expected to under perform the tree-based models on those predictors' contribution; this is broadly consistent with the random forest outperforming logistic regression in every phase (Table 1), though this is not a formal diagnostic and other explanations cannot be ruled out.

AUC was used as the primary comparison metric, following Ryan et al. (2025), for two reasons. First, unlike sensitivity, specificity, and accuracy, AUC does not depend on an arbitrarily chosen classification threshold; it summarises a model's ability to rank compliant respondents above non-compliant ones across all possible thresholds at once. Second, using the same primary metric as Ryan et al. (2025) allows this project's baseline results to be compared directly against their published benchmarks as a check on the pipeline (Results).

6 Results

6.1 Descriptive Statistics

The analytical sample ($n = 41{,}168$) had a mean age of 48.0 years (SD 17.1). Survey weeks ranged from 6 to 51. Overall mask compliance was 52.1% ($n = 21{,}468$ compliant).

Figure 2 shows mask compliance by state. Victoria had the highest compliance at 75.3%, while Tasmania had the lowest at 12.3%, a gap of 63.0 percentage points.

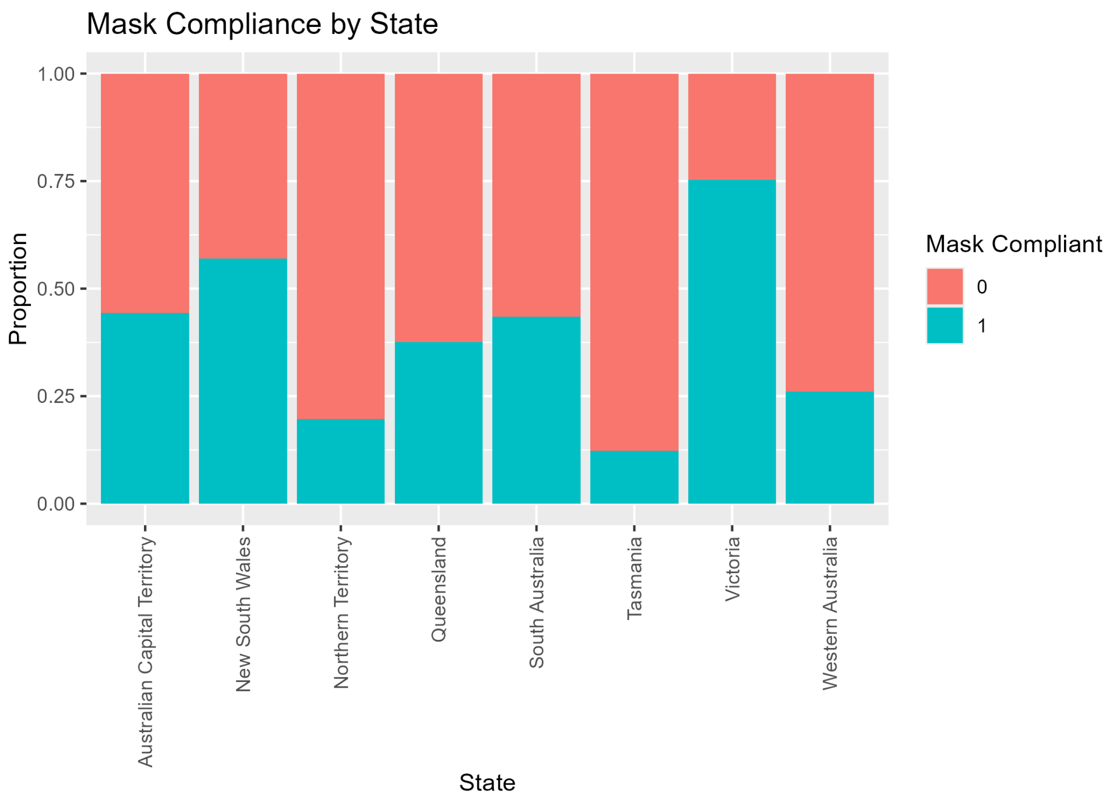

Figure 2: Proportion of respondents classified as mask-compliant (scoring ≥ 4 across four mask-wearing scenarios) by Australian state.

Figure 3 shows that compliance rose substantially over the survey period, particularly from mid-2020 onward.

Figure 4 shows the relationship between the behaviour score and mask compliance.

6.2 Model Performance

Table 1 reports all model performance metrics on the held-out test set ($n = 12{,}351$ for each model).

6.3 Predictor Significance

Table 2 reports statistical significance tests for each predictor in the three logistic regression models.

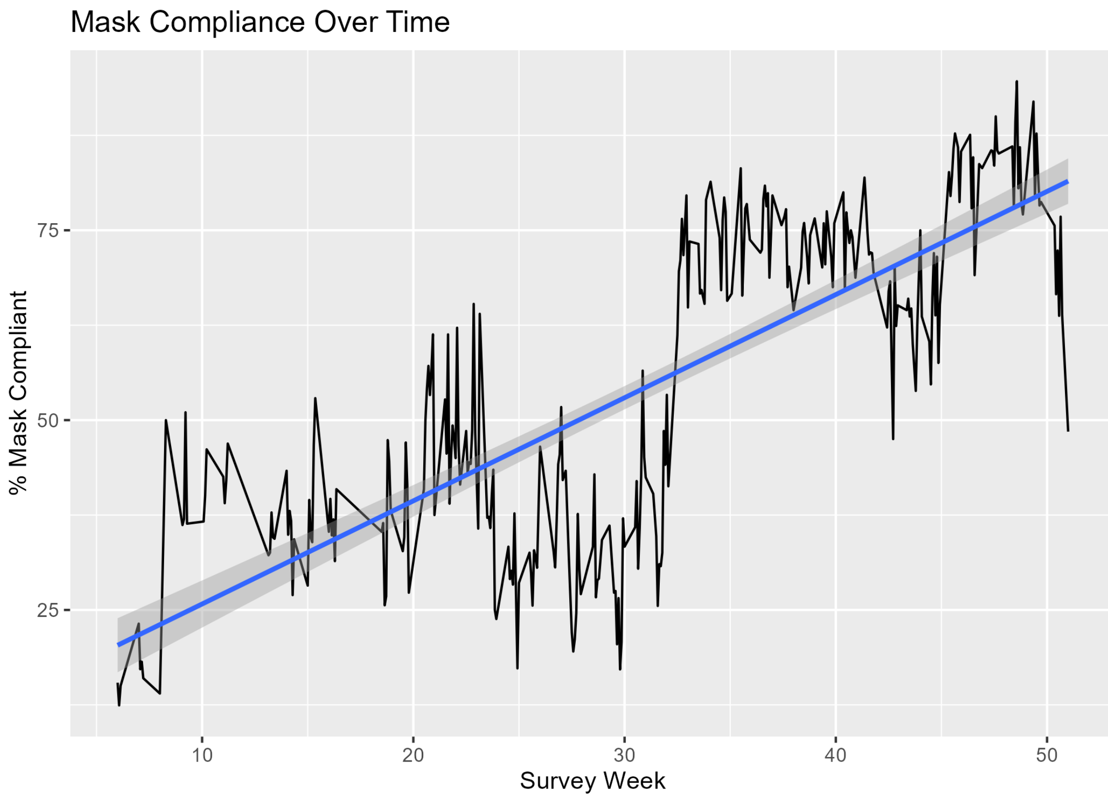

Figure 3: Estimated percentage of mask-compliant respondents by survey fortnight (fortnights with fewer than 30 respondents excluded). Note: data begin at week 6 because the relevant survey items were not introduced until that wave.

Model Dataset	Model	AUC	Sens.	Spec.	Acc.
No State	Logistic Regression	0.824	0.770	0.718	0.745
No State	Classification Tree	0.810	0.807	0.699	0.756
No State	Random Forest	0.854	0.794	0.748	0.772
With State	Logistic Regression	0.860	0.804	0.758	0.782
With State	Classification Tree	0.806	0.783	0.778	0.781
With State	Random Forest	0.906	0.853	0.806	0.831
Measurable Factors	Logistic Regression	0.860	0.807	0.747	0.779
Measurable Factors	Classification Tree	0.779	0.789	0.726	0.759
Measurable Factors	Random Forest	0.911	0.863	0.813	0.839

Table 1: Model performance on the held-out test set. Sens. = sensitivity, Spec. = specificity, Acc. = accuracy. Each model dataset uses its own independent 70/30 split, using the same random seed in every case.

6.4 No-State Models

Figures 5, 6, and 7 show the ROC curves for the no-state logistic regression (AUC = 0.824), classification tree (AUC = 0.810), and random forest (AUC = 0.854). Figure 8 shows random forest variable importance.

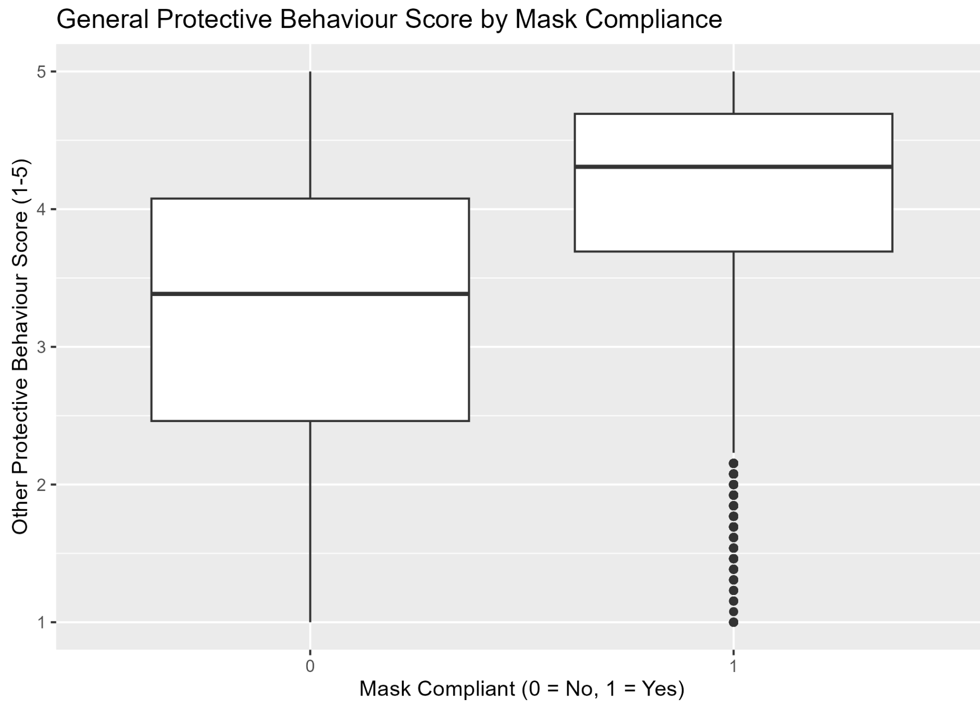

Figure 4: Distribution of behaviour score (mean of 13 behaviour items, 1–5 scale) by mask compliance status.

Predictor	No State		With State		Measurable Factors	
	χ^2	p	χ^2	p	χ^2	p
Survey week	4625.9	<0.001	5095.4	<0.001	2605.9	<0.001
State of residence	–	–	2964.2	<0.001	–	–
Age	158.9	<0.001	134.3	<0.001	131.7	<0.001
Gender	19.1	<0.001	9.4	0.002	8.9	0.003
Household size	2.0	0.156	3.7	0.056	4.5	0.034
Employment status	74.6	<0.001	55.4	<0.001	56.3	<0.001
Wellbeing	64.3	<0.001	73.5	<0.001	89.7	<0.001
Perceived severity	86.5	<0.001	135.6	<0.001	120.1	<0.001
Willingness to self-isolate	19.9	<0.001	63.7	<0.001	42.4	<0.001
Mental health response status	3.1	0.208	1.1	0.588	24.9	<0.001
General protective-behaviour score	5081.5	<0.001	3746.9	<0.001	3586.2	<0.001
Face-covering policy strictness	–	–	–	–	665.6	<0.001
Stay-at-home policy strictness	–	–	–	–	1.8	0.175
Median weekly household income	–	–	–	–	232.0	<0.001
Percentage with Bachelor's or higher	–	–	–	–	447.6	<0.001
Percentage born overseas	–	–	–	–	111.0	<0.001

Table 2: Statistical significance (χ^2 test) of each predictor in the three logistic regression models. Degrees of freedom omitted for brevity (1 for all rows except state, df = 7, and employment status, df = 4).

6.5 Models With State Included

Figures 9, 10, and 11 show the ROC curves for the with-state models (logistic regression AUC = 0.860, classification tree AUC = 0.806, random forest AUC = 0.906). Figure 12 shows variable importance with state included.

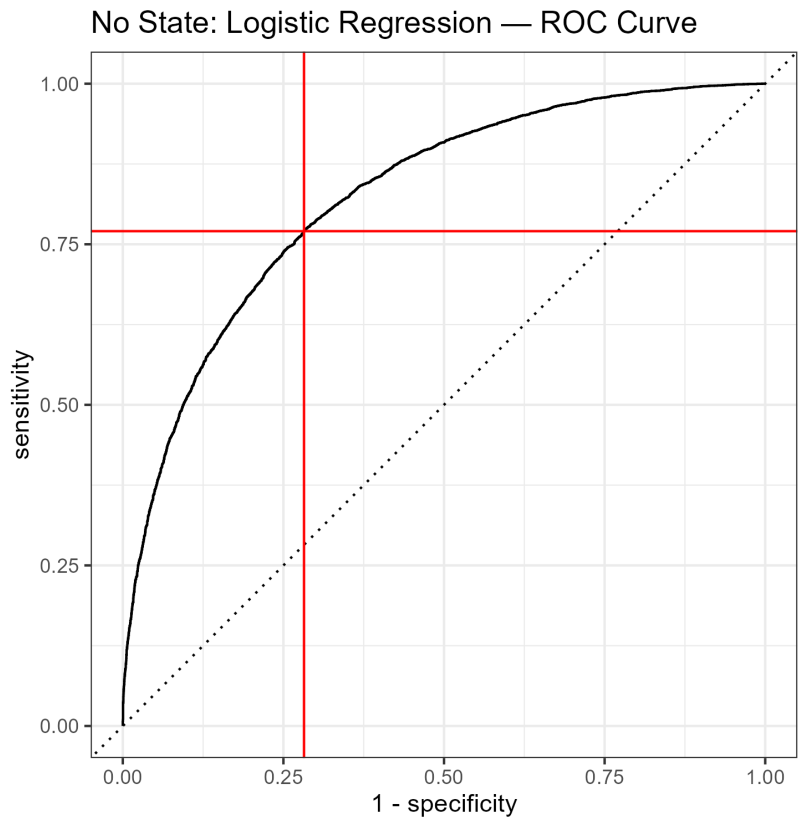

Figure 5: ROC curve for the no-state logistic regression model (AUC = 0.824). Red lines mark sensitivity (0.770) and 1 − specificity (0.282) at the 0.5 classification threshold.

6.6 Models With Measurable State Predictors

Figures 13, 14, and 15 show the ROC curves for the measurable-factors models (logistic regression AUC = 0.860, classification tree AUC = 0.779, random forest AUC = 0.911). Figures 16 and 17 show variable importance for the random forest and classification tree respectively.

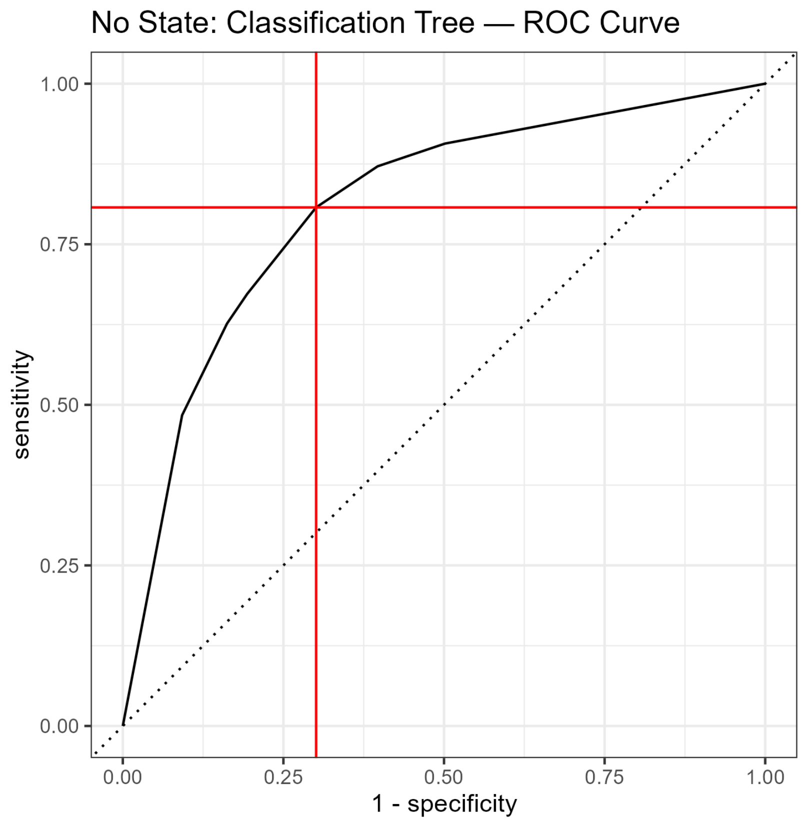

Figure 6: ROC curve for the no-state classification tree model (AUC = 0.810). Red lines mark sensitivity (0.807) and 1 – specificity (0.301) at the 0.5 threshold.

7 Discussion

7.1 The State Effect Exists

The no-state random forest achieved AUC = 0.854 (Figure 7) using only individual-level predictors. As shown in Figure 8, survey week and behaviour score dominate. No single individual-level characteristic strongly predicts compliance on its own. Adding state as a predictor increased AUC to 0.906 (Figure 11), a gain of 0.052 AUC points, confirming the finding of Ryan et al. (2025) that state carries meaningful information beyond what individuals alone can explain.

7.2 The State Effect Is Explainable

Replacing state with five measurable state-level predictors achieved AUC = 0.911 (Figure 15), matching or slightly exceeding the with-state model (0.906). As shown in Figure 16, mask-policy strictness ranks near the top of the resulting importance ranking, close to the position state occupied when included directly (Figure 12). This indicates that policy stringency is the primary mechanism through which state differences in compliance arise. States with stricter face-covering requirements had higher compliance, and this relationship can now be quantified rather than attributed to an unexplainable label.

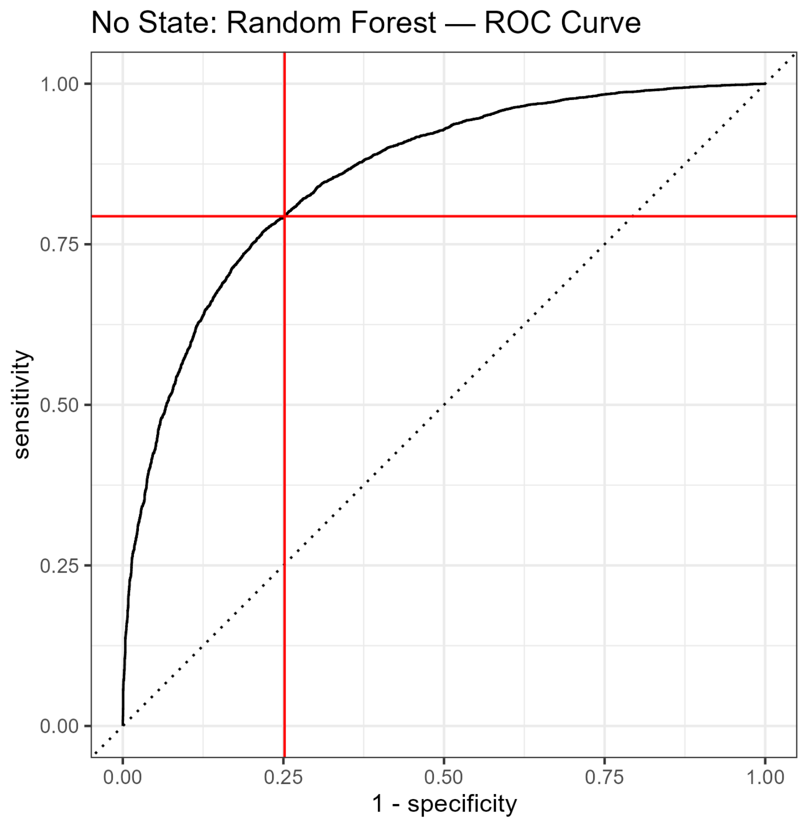

Figure 7: ROC curve for the no-state random forest model (AUC = 0.854). Red lines mark sensitivity (0.794) and 1 − specificity (0.252) at the 0.5 threshold.

The remaining measurable predictors (lockdown strictness, education, overseas-born proportion, income) contribute additional but smaller signal, suggesting that demographic composition and a second, distinct policy lever play a secondary role alongside face-covering policy specifically.

7.3 Why Compliance Differed by State and Rose Over Time

The two clearest descriptive patterns in this project, the gap between states in Figure 2 and the rise over time in Figure 3, line up closely with a small number of dated, real mandate and outbreak events. Tracing them out helps explain why mask-policy strictness carries so much of the predictive weight that state itself carried when included directly.

Victoria, the highest-compliance state in Figure 2 at 75.3%, is the state described in the Introduction as the first to mandate mask-wearing, from 23 July 2020 (Costantino and Raina MacIntyre 2021, 1). This gives it one of the longest cumulative periods under an active mandate of any state, consistent with its position at the top of Figure 2. Tasmania, the lowest-compliance state at 12.3%, recorded comparatively few cases throughout the survey period and did not experience a comparable sustained-mandate outbreak. This is consistent with the same mechanism, though it was not independently checked against a specific policy-history source and should be read as plausible rather than confirmed.

The rise in Figure 3 coincides with survey week 31, the week in which the Delta variant was first detected in Sydney's eastern suburbs on 16 June 2021, forcing New South Wales into mandatory

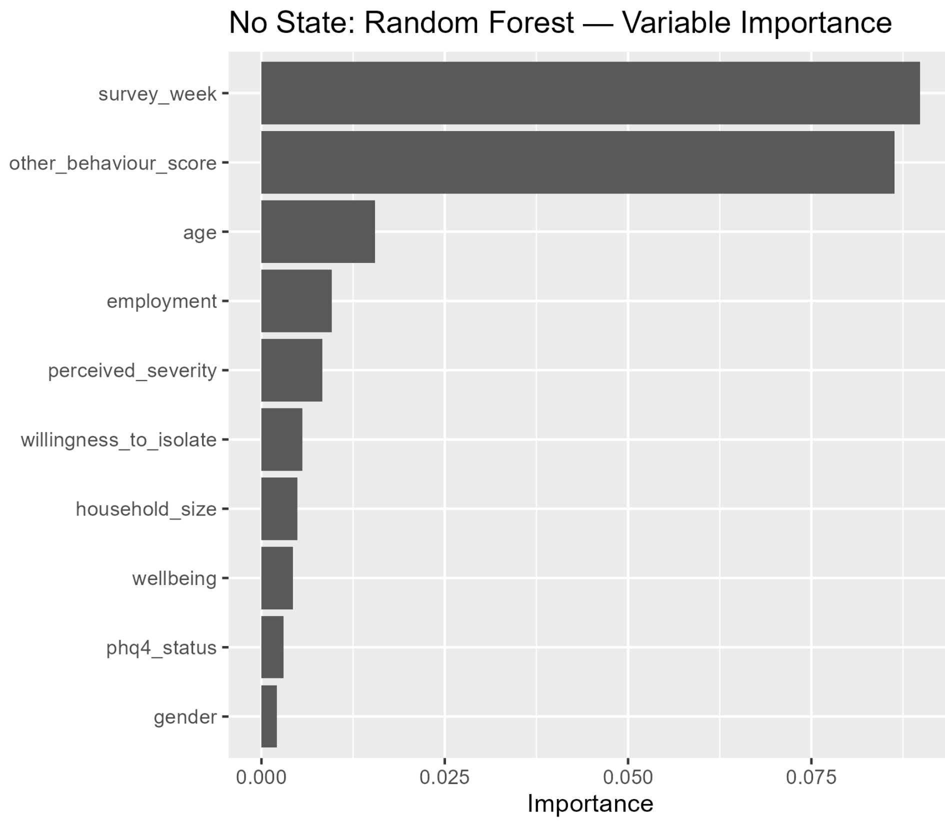

Figure 8: Permutation variable importance for the no-state random forest model. Survey week and behaviour score dominate, with all remaining predictors substantially lower.

masks within days, a Greater Sydney lockdown from 25 June, and a state-wide lockdown from 15 August 2021 (National Centre for Immunisation Research and Surveillance (NCIRS) 2021). This cannot be attributed to vaccination, worth stating explicitly since vaccination is often assumed to explain rising 2021 compliance. National full-vaccination coverage was below 8% in early July 2021 (The Guardian Australia 2021) and only around 26% by mid-August (Voice of America News 2021), well after this rise appears.

Vaccination becomes relevant later in 2021, but in the opposite direction. The federal National Plan tied the *relaxation* of restrictions, not their tightening, to vaccination thresholds of 70% and 80% (Bloomberg 2021). When New South Wales passed 80% on 18 October 2021, survey week 40, masks were explicitly lifted in some indoor settings (Inquirer.net 2021). That easing reversed when Omicron drove case numbers to record highs, and both states reinstated masks on 23–24 December 2021, survey week 45, the event described in the Introduction, with national double-dose coverage by then above 90% (Gulf News 2021). The highest vaccination point in the whole survey period is therefore also the point masks were just reinstated after being relaxed. This is the opposite of a vaccination-driven story, and is consistent instead with mask-policy strictness being the real mechanism.

Week numbers above are computed precisely from each event's calendar date using the survey-week formula in Methods.

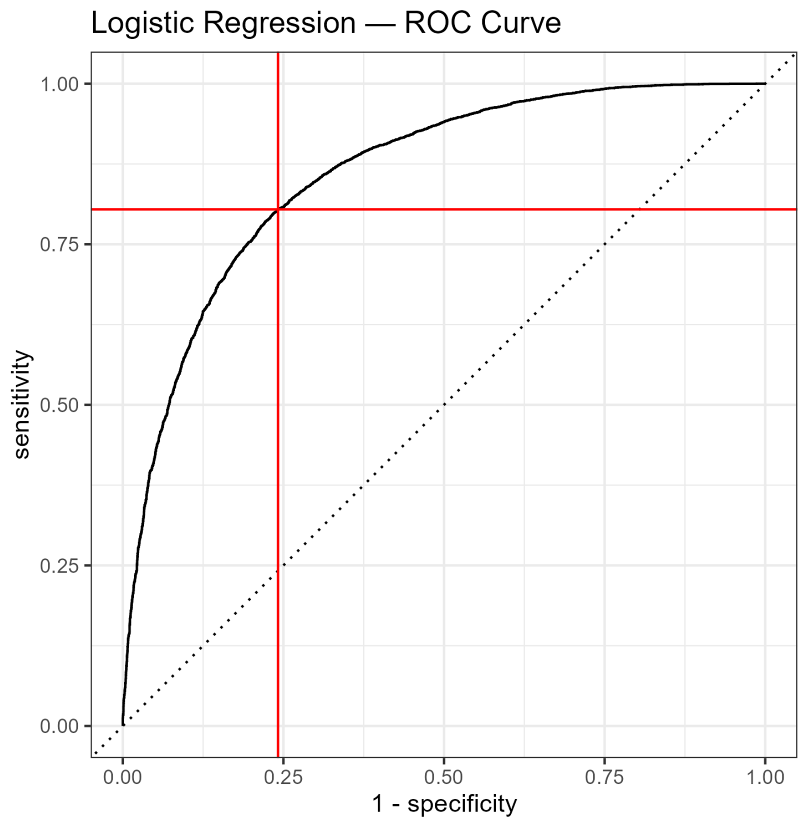

Figure 9: ROC curve for the with-state logistic regression model (AUC = 0.860). Red lines mark sensitivity (0.804) and 1 – specificity (0.242) at the 0.5 threshold.

7.4 The Classification Tree Underperforms, and Gets Worse With Measurable Predictors

The classification tree is the weakest of the three models throughout (Table 1), and its behaviour across the three model datasets is itself informative. Its AUC falls from 0.810 (No State, Figure 6) to 0.806 (With State, Figure 10) to 0.779 (Measurable Factors, Figure 14), a further decline once state is replaced by measurable predictors, whereas logistic regression stays essentially flat (0.824, 0.860, 0.860; Figures 5, 9, 13) and random forest improves over the same change. A single tree makes a clean either/or split on a categorical variable like state, which it exploits directly, as illustrated in Background (Figure 1). Mask-policy strictness and lockdown strictness are both continuous, and a tree must commit to a single threshold at each split, discarding information a linear model or a forest does not discard.

Figure 17 makes this concrete. Lockdown strictness does not appear in the measurable-factors tree's importance ranking at all, because the tree's splits never used it, whereas the equivalent random forest ranking (Figure 16) places it seventh of ten, a modest but genuine contribution the single tree could not find. This matches lockdown strictness contributing no significant improvement to the measurable-factors logistic regression's fit either, with a significance value of 1.8 ($p = 0.17$) against 665.6 ($p < 2 \times 10^{-16}$) for face-covering strictness (Table 2). The two policy variables are correlated over time, since states that locked down harder also mandated masks harder, so a model considering one predictor's contribution at a time finds little left for the second to

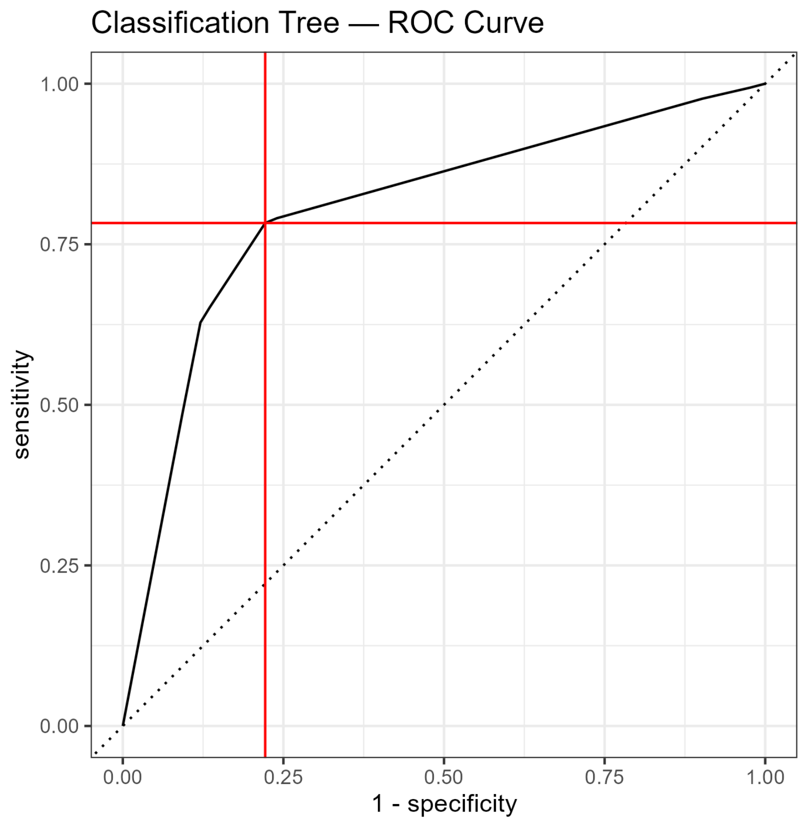

Figure 10: ROC curve for the with-state classification tree model (AUC = 0.806). Red lines mark sensitivity (0.783) and 1 – specificity (0.222) at the 0.5 threshold.

explain, while a model that can exploit interactions still finds some independent signal in it.

7.5 What the Individual-Level Predictors Show

Table 2 shows that most individual-level predictors, survey week, age, employment status, wellbeing, perceived severity, willingness to self-isolate, and behaviour score, were statistically significant across all three model datasets (all $p < 0.05$), indicating their relationship with mask compliance is not an artefact of whichever other predictors happen to be in the model. Behaviour score's relationship with compliance is shown directly in Figure 4, where compliant respondents have a visibly higher median score than non-compliant respondents, consistent with its consistently high ranking across every variable importance figure in this report. Two predictors are less consistent. Household size is significant only in the measurable-factors model ($p = 0.034$). Mental health response status is non-significant in the No State and With State models but becomes clearly significant once state is replaced by measurable predictors ($p < 0.001$), suggesting state, as a fixed-effect category, was partly absorbing variation that mental health status would otherwise explain.

7.6 Limitations

The ABS 2021 Census indicators are a single cross-sectional snapshot and cannot capture within-state demographic change across the two-year study window. The four high-missingness variables

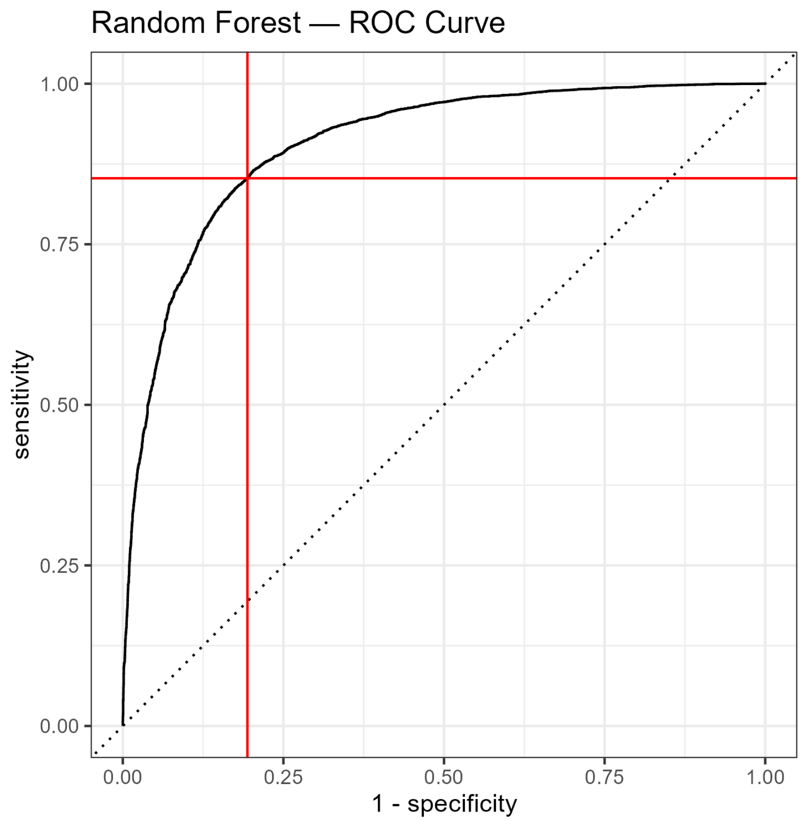

Figure 11: ROC curve for the with-state random forest model (AUC = 0.906). Red lines mark sensitivity (0.853) and 1 – specificity (0.194) at the 0.5 threshold.

excluded from modelling, particularly non-household contacts, a consistent top predictor in Ryan et al. (2025), represent real information loss. State-level indicators are ecological variables, and causal inference at the individual level is not supported. The real-world events described above are a plausible, dated account consistent with the models' own findings, not an independent causal test, and other unmeasured events occurring around the same weeks cannot be ruled out as contributing causes.

7.7 Next Steps

The most direct extension is a formal mediation or sensitivity analysis of the behaviour score's role in the policy effect identified above, noted as unresolved. A second is testing whether the two policy indicators' shared time trend can be separated more directly, for example by statistically removing the part of stay-at-home strictness that overlaps with face-covering strictness before fitting, to determine whether the classification tree and random forest results reflect genuine independent signal in stay-at-home policy or simply a nonlinear use of the time trend face-covering strictness already captures.

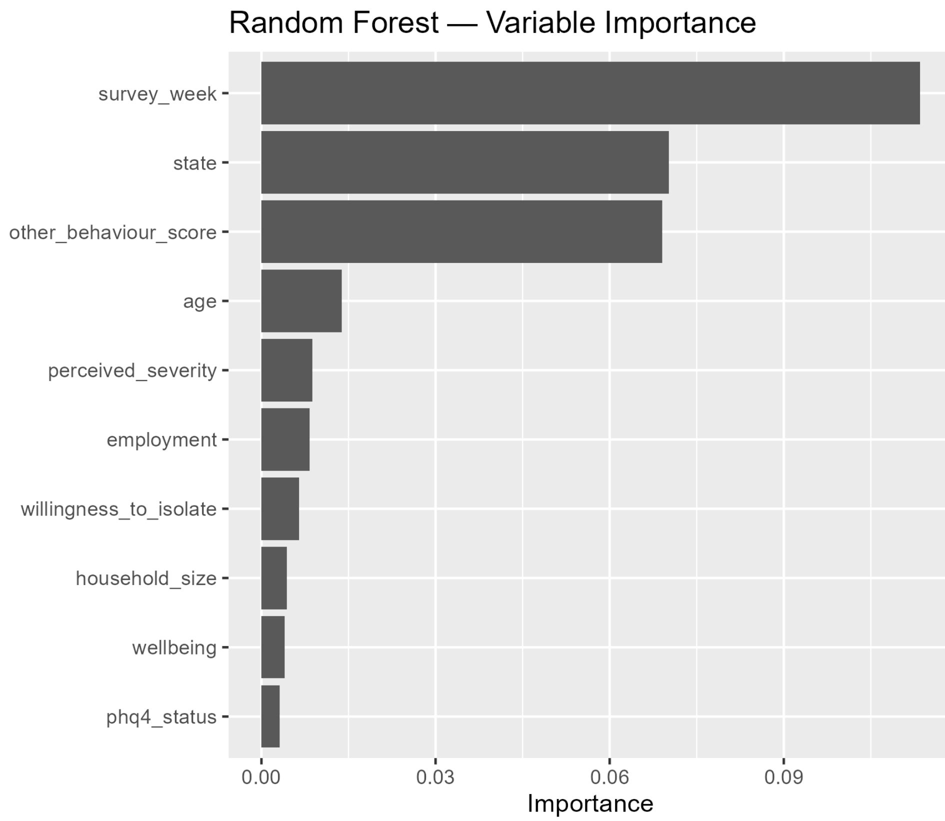

Figure 12: Permutation variable importance for the with-state random forest model. Survey week ranks first, behaviour score second, and state third.

7.8 Conclusion

This project set out to test whether the state effect identified by Ryan et al. (2025), in which state was among the most important predictors of mask-wearing compliance in their models, could be explained by measurable factors rather than treated as an unexplained label. Replacing state with government policy indicators and Census demographic indicators matched the performance achieved with state included directly, and mask-policy strictness emerged as the clear replacement for what state had been doing. This finding lines up closely with real, dated mask-mandate and outbreak events across the survey period, from Victoria's first mandate in 2020 through the Delta-driven rise in mid-2021 to the Omicron-driven reinstatement of masks in December 2021. The state effect therefore appears substantially explainable rather than mysterious, though not fully so. A modest performance gap between the with-state and measurable-factors models remains, and slower-moving factors such as regional demographic composition are harder to verify against a specific timeline than the mandate-driven policy story is. The classification tree's consistent under performance, and its particular difficulty with continuous policy predictors, is a secondary but genuine finding of its own, relevant to future algorithm choice for similar state-comparison problems.

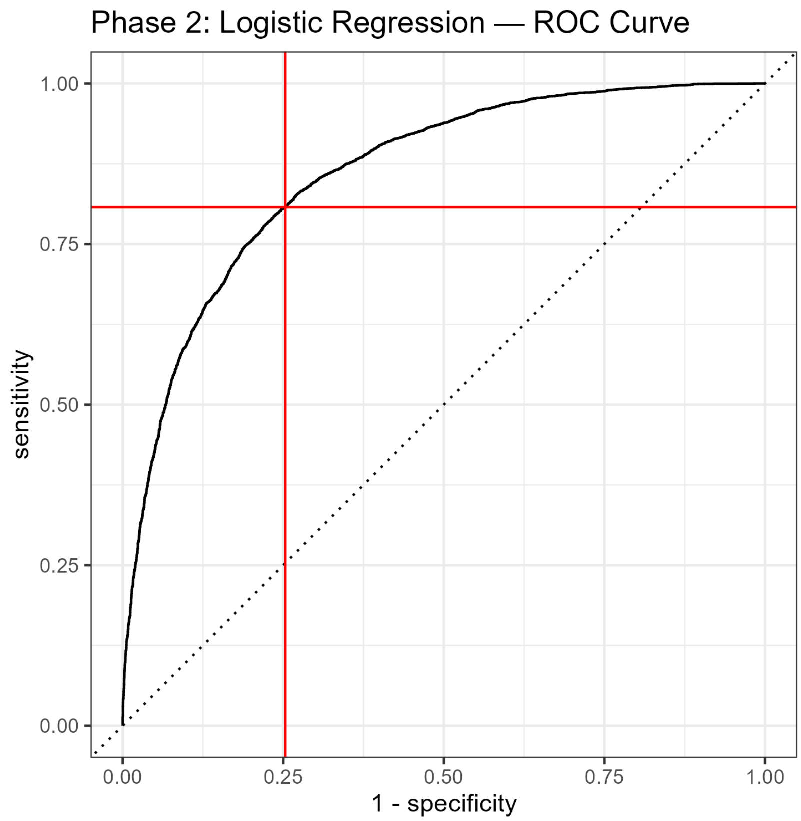

Figure 13: ROC curve for the measurable-factors logistic regression model (AUC = 0.860). State replaced by five measurable state-level predictors.

8 Acknowledgements

I would like to thank my supervisor, Anthony Mays, for his guidance and support throughout this project.I would also like to thank Matthew Ryan, whose paper (Ryan et al. 2025) this project extends.

Course materials from MATHS 7107 Data Taming, School of Mathematical Sciences, University of Adelaide, were used for methodological guidance, including the logistic regression formulation and the AUC interpretation scale used in Background (University of Adelaide, School of Mathematical Sciences 2024a), and the classification tree node terminology used in Background (University of Adelaide, School of Mathematical Sciences 2024b).

I acknowledge the use of AI tools for debugging code and for grammar and clarity checks on the written report. All modelling decisions, analysis, and interpretation are my own.

The code and data underlying this project are available in a public repository at https://github.com/myashkumar/MDSresarch_FinalReport_Mahesh_Y

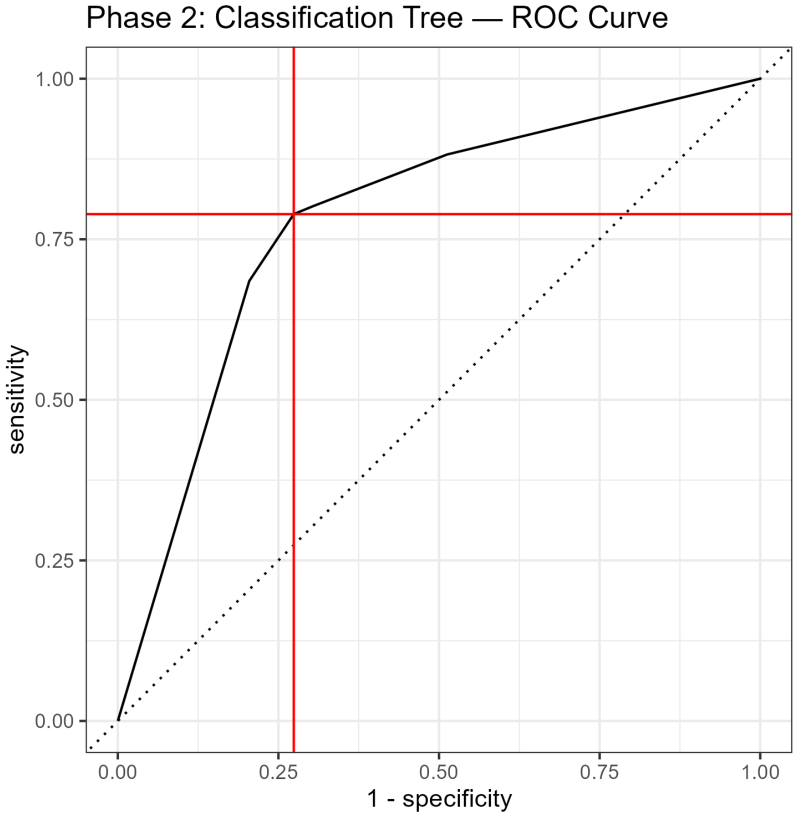

Figure 14: ROC curve for the measurable-factors classification tree model (AUC = 0.779).

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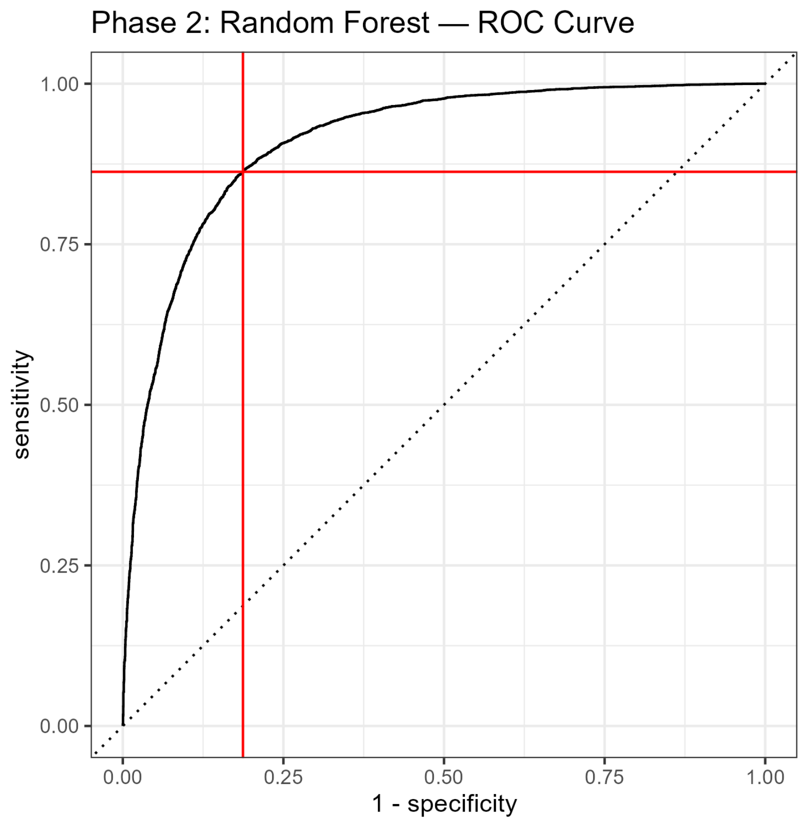

Figure 15: ROC curve for the measurable-factors random forest model (AUC = 0.911). State replaced by five measurable state-level predictors.

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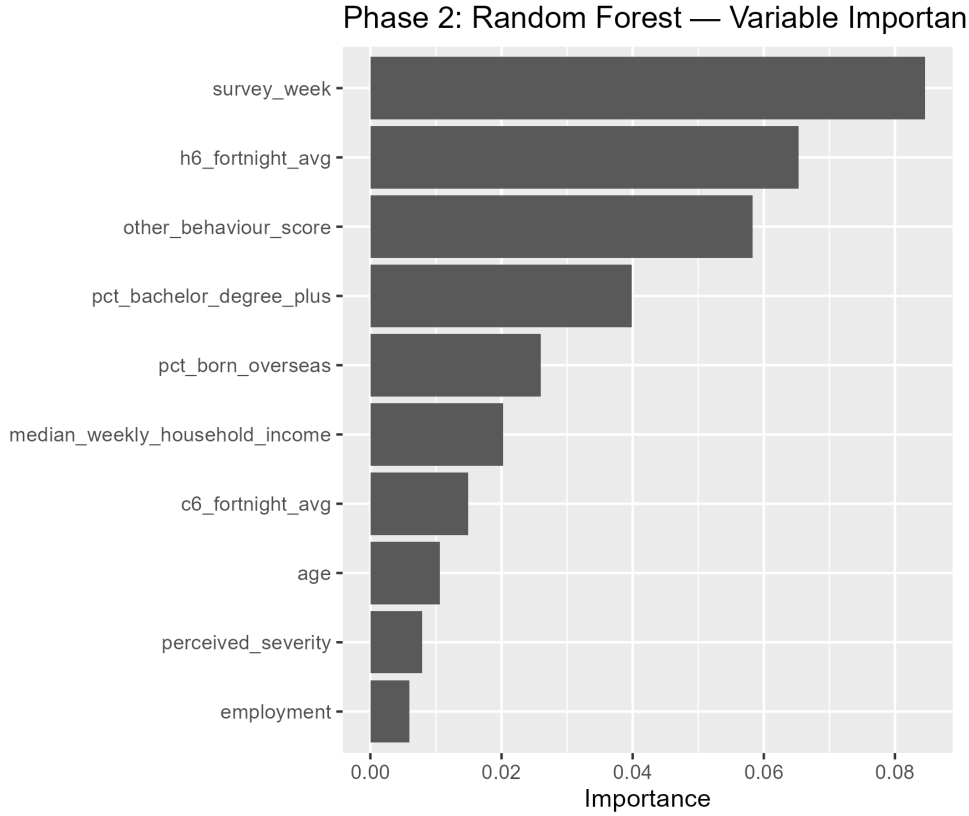

Figure 16: Permutation variable importance for the measurable-factors random forest model. Mask-policy strictness ranks near the top, replacing state from the with-state model. Lockdown strictness contributes a smaller but non-trivial amount.

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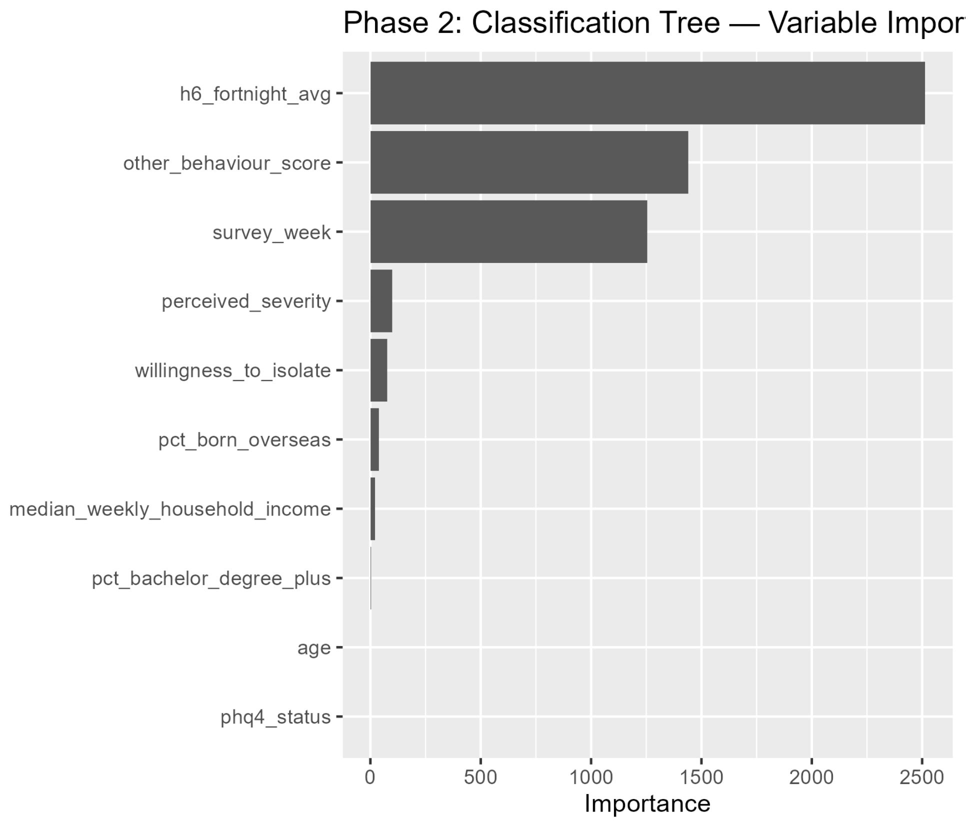

Figure 17: Permutation variable importance for the measurable-factors classification tree model. Mask-policy strictness ranks first. Lockdown strictness does not appear at all, since the tree's splits (Appendix B) never used it.

10 Statistical Appendix

All results in this report were generated by a single R script combining the cleaning pipeline and model fitting (documented separately below as Appendix A and Appendix B for clarity). The script requires `australia.csv`, `OxCGRT_AUS_latest.csv`, and `abs_state_indicators_2021.csv` in the working directory.

10.1 Appendix A: Data Loading and Cleaning

The three source files are read in directly.

```r
raw            <- read_csv("australia.csv")
oxcgrt_raw     <- read_csv("OxCGRT_AUS_latest.csv")
abs_indicators <- read_csv("abs_state_indicators_2021.csv")
```

Character columns encoding missing values as a blank space are converted to a proper missing-value code.

```r
df <- mutate(df,
             gender = ifelse(str_replace_all(gender, " ", "") == "",
                             NA_character_, gender),
             state  = ifelse(str_replace_all(state, " ", "") == "",
                             NA_character_, state))
```

Numeric variables stored as text are converted to numbers, and out-of-range values on the two 0–10 scales are treated as missing.

```r
df <- mutate(df,
             r1_1  = as.numeric(str_match(r1_1, "(\\d+)")[,2]),
             age   = as.numeric(age))

df_final <- mutate(df_final,
                   govt_handling = ifelse(govt_handling > 10, NA,
                                          govt_handling),
                   perceived_susceptibility = ifelse(
                     perceived_susceptibility > 10, NA,
                     perceived_susceptibility))
```

All Likert-scale survey items are recoded from text labels to integers using this pattern.

```r
df <- mutate(df,
             m1_1_num = ifelse(i12_health_1 == "Not at all", 1,
                        ifelse(i12_health_1 == "Rarely", 2,
                        ifelse(i12_health_1 == "Sometimes", 3,
                        ifelse(i12_health_1 == "Frequently", 4,
                        ifelse(i12_health_1 == "Always", 5, NA_real_))))))
```

Household size is capped and recoded, and employment status is collapsed into five categories by matching key phrases in the raw text.

```r
df <- mutate(df,
  household_size_num = ifelse(
    household_size == "8 or more", 8,
    ifelse(household_size %in% c("Prefer not to say", "Don't know"),
           NA_real_, as.numeric(household_size))),
  employment_clean = ifelse(
    str_detect(employment_status, "ull time"), "Full time",
    ifelse(str_detect(employment_status, "art time"), "Part time",
    ifelse(str_detect(employment_status, "Retired"), "Retired",
    ifelse(str_detect(employment_status, "Unemployed"), "Unemployed",
    ifelse(str_detect(employment_status, "Not working"), "Not working",
           "Other"))))),
  employment_clean = as.factor(employment_clean))
```

The mask-compliance outcome and the general behaviour score are built by averaging their respective survey items and thresholding at 4.

```
df <- mutate(df,
             face_mask_score = (m1_1_num + m1_2_num + m1_3_num + m1_4_num) / 4,
             face_mask_wearing = ifelse(face_mask_score >= 4, 1, 0),
             other_behaviour_score = (m9_1_num + m9_2_num + m9_3_num +
                                      m9_4_num + m9_5_num + m9_6_num +
                                      m9_7_num + m9_8_num + m9_9_num +
                                      m9_10_num + m9_11_num + m9_12_num +
                                      m9_13_num) / 13)
mean(df$face_mask_wearing, na.rm = TRUE) * 100
```

The output confirms overall compliance of 52.1%.

The response timestamp is parsed into a date, and survey week is defined as the number of complete two-week periods elapsed since 1 April 2020.

```
df <- mutate(df,
             date_part   = str_match(endtime, "(\\d{2}/\\d{2}/\\d{4})")[,2],
             survey_date = dmy(date_part),
             survey_week = as.integer(survey_date -
                             min(survey_date, na.rm = TRUE)) / 14)
```

Each respondent is assigned a mental health response status based on whether the PHQ-4 module was answered, declined, or left blank.

```
df <- mutate(df,
             phq4_status = ifelse(phq4_declined, "Declined",
                            ifelse(phq4_blank, "Not available", "Answered")),
             phq4_status = as.factor(phq4_status))
```

The output confirms 41,568 Answered, 1,294 Declined, and 10,971 Not available.

The four high-missingness variables are dropped from the dataset.

```
df_final <- select(df_final, -govt_handling, -perceived_susceptibility,
                   -ease_of_isolation, -contacts)
```

Respondents missing any modelling variable are removed, giving the final analytical sample.

```
df_clean <- filter(df,
                   !is.na(face_mask_wearing) & !is.na(general_behaviour) &
                   !is.na(age) & !is.na(gender) & !is.na(state) &
                   !is.na(household_size_num) & !is.na(employment_clean) &
                   !is.na(cantril_ladder) & !is.na(i11_health_num) &
                   !is.na(survey_week) & !is.na(other_behaviour_score))
nrow(df_clean)
```

The output confirms 41,168 respondents retained from 53,833.

The three model datasets are built by selecting the relevant predictors for each and removing any remaining incomplete rows.

```
df_model_no_state <- df_final %>%
  select(-perceived_susceptibility, -ease_of_isolation,
         -govt_handling, -contacts) %>%
  mutate(face_mask_wearing = as.factor(
    ifelse(face_mask_wearing == 1, "yes", "no"))) %>%
  select(face_mask_wearing, survey_week, age, gender, household_size,
         employment, wellbeing, perceived_severity,
         willingness_to_isolate, phq4_status,
         other_behaviour_score) %>%
  filter(if_all(everything(), ~ !is.na(.)))

df_model <- df_final %>%
  select(-perceived_susceptibility, -ease_of_isolation,
         -govt_handling, -contacts) %>%
  mutate(face_mask_wearing = as.factor(
    ifelse(face_mask_wearing == 1, "yes", "no"))) %>%
  select(face_mask_wearing, survey_week, state, age, gender,
         household_size, employment, wellbeing, perceived_severity,
         willingness_to_isolate, phq4_status,
         other_behaviour_score) %>%
  filter(if_all(everything(), ~ !is.na(.)))

df_model_phase2 <- df_final_p2 %>%
  select(-perceived_susceptibility, -ease_of_isolation,
         -govt_handling, -contacts, -state) %>%
  mutate(face_mask_wearing = as.factor(
    ifelse(face_mask_wearing == 1, "yes", "no"))) %>%
  select(face_mask_wearing, survey_week, age, gender, household_size,
         employment, wellbeing, perceived_severity,
         willingness_to_isolate, phq4_status, other_behaviour_score,
         h6_fortnight_avg, c6_fortnight_avg,
         median_weekly_household_income, pct_bachelor_degree_plus,
         pct_born_overseas) %>%
  filter(if_all(everything(), ~ !is.na(.)))
```

State-level rows are extracted from the daily policy tracker, averaged within each state and survey fortnight, and merged onto each respondent alongside the Census demographic indicators.

```
oxcgrt <- filter(oxcgrt_raw, Jurisdiction == "STATE_TOTAL")
oxcgrt <- mutate(oxcgrt,
                 state           = RegionName,
                 ox_date         = ymd(Date),
```

```r
                  h6              = `H6M_Facial Coverings`,
                  c6              = `C6M_Stay at home requirements`,
                  fortnight_index = as.integer(as.integer(
                    ox_date - min(df_final$survey_date,
                                   na.rm = TRUE)) / 14))

oxcgrt_avg <- oxcgrt %>%
  group_by(state, fortnight_index) %>%
  summarise(h6_fortnight_avg = mean(h6, na.rm = TRUE),
            c6_fortnight_avg = mean(c6, na.rm = TRUE))

df_final_p2 <- mutate(df_final,
                      fortnight_index = as.integer(as.integer(survey_date -
                                          min(survey_date, na.rm = TRUE)) / 14))
df_final_p2 <- left_join(df_final_p2, oxcgrt_avg,
                          by = c("state", "fortnight_index"))
df_final_p2 <- left_join(df_final_p2, abs_indicators, by = "state")
```

10.2 Appendix B: Model Fitting and Evaluation

Each of the three model datasets is split independently, with the same seed set immediately before every split.

```r
set.seed(1914220)
df_split_ns <- initial_split(df_model_no_state, prop = 0.7)
df_train_no_state <- training(df_split_ns)
df_test_no_state  <- testing(df_split_ns)

set.seed(1914220)
df_split <- initial_split(df_model, prop = 0.7)
df_train <- training(df_split)
df_test  <- testing(df_split)

set.seed(1914220)
df_split_p2 <- initial_split(df_model_phase2, prop = 0.7)
df_train_p2 <- training(df_split_p2)
df_test_p2  <- testing(df_split_p2)
```

The three No State models are fitted first, achieving AUC 0.824, 0.810, and 0.854 respectively.

```r
lr_ns <- logistic_reg() %>%
  set_engine("glm") %>%
  fit(face_mask_wearing ~ ., data = df_train_no_state)

ct_ns <- decision_tree(mode = "classification") %>%
```

```
  set_engine("rpart") %>%
  fit(face_mask_wearing ~ ., data = df_train_no_state)

rf_ns <- rand_forest(mode = "classification") %>%
  set_engine("ranger", importance = "permutation") %>%
  fit(face_mask_wearing ~ ., data = df_train_no_state)
```

The three With-State models follow the same pattern, achieving AUC 0.860, 0.806, and 0.906.

```
lrfit <- logistic_reg() %>%
  set_engine("glm") %>%
  fit(face_mask_wearing ~ ., data = df_train)

ctree <- decision_tree(mode = "classification") %>%
  set_engine("rpart") %>%
  fit(face_mask_wearing ~ ., data = df_train)

rf <- rand_forest(mode = "classification") %>%
  set_engine("ranger", importance = "permutation") %>%
  fit(face_mask_wearing ~ ., data = df_train)
```

The three Measurable-Factors models complete the set, achieving AUC 0.860, 0.779, and 0.911.

```
lr_p2 <- logistic_reg() %>%
  set_engine("glm") %>%
  fit(face_mask_wearing ~ ., data = df_train_p2)

ct_p2 <- decision_tree(mode = "classification") %>%
  set_engine("rpart") %>%
  fit(face_mask_wearing ~ ., data = df_train_p2)

rf_p2 <- rand_forest(mode = "classification") %>%
  set_engine("ranger", importance = "permutation") %>%
  fit(face_mask_wearing ~ ., data = df_train_p2)
```

Predictor significance for the Measurable-Factors logistic regression is obtained with a Type II test.

```
Anova(lr_p2$fit)
```

The output gives the values reported in Table 2: face-covering strictness $\chi^2 = 665.6$ ($p < 2.2 \times 10^{-16}$), median household income $\chi^2 = 232.0$ ($p < 2.2 \times 10^{-16}$), Bachelor's-degree percentage $\chi^2 = 447.6$ ($p < 2.2 \times 10^{-16}$), overseas-born percentage $\chi^2 = 111.0$ ($p < 2.2 \times 10^{-16}$), and lockdown strictness $\chi^2 = 1.8$ ($p = 0.1745$).

The performance of all nine models is compared in a single summary table.

```r
tibble(
  `Model Dataset` = rep(c("No State", "With State",
                          "Measurable Factors"), each = 3),
  Model = rep(c("Logistic Regression", "Classification Tree",
                "Random Forest"), times = 3),
  AUC = c(0.824, 0.810, 0.854, 0.860, 0.806, 0.906,
          0.860, 0.779, 0.911),
  Sensitivity = c(0.770, 0.807, 0.794, 0.804, 0.783, 0.853,
                  0.807, 0.789, 0.863),
  Specificity = c(0.718, 0.699, 0.748, 0.758, 0.778, 0.806,
                  0.747, 0.726, 0.813),
  Accuracy = c(0.745, 0.756, 0.772, 0.782, 0.781, 0.831,
               0.779, 0.759, 0.839)
)
```

The resulting table matches Table 1 exactly.

10.3 Appendix C: Measurable-Factors Merge Validation

The row count and missingness of the merged dataset are checked to confirm the merge introduced no lost rows or missing values.

```r
nrow(df_final)
nrow(df_final_p2)
sum(is.na(df_final_p2$h6_fortnight_avg))
sum(is.na(df_final_p2$c6_fortnight_avg))
sum(is.na(df_final_p2$median_weekly_household_income))
```

The output confirms 41,168 rows in both datasets, with zero missing values in each of the three checked columns.

10.4 Appendix D: ABS DataPack Derivation and Correction

`abs_state_indicators_2021.csv` was derived from three ABS 2021 Census DataPack tables at State/Territory geography: `2021Census_G01_AUST_STE.csv`, `2021Census_G02_AUST_STE.csv`, and `2021Census_G49B_AUST_STE.csv`. `population_2021` and `median_weekly_household_income` were checked against these tables and matched exactly for all eight states/territories with no adjustment required. During verification, `pct_born_overseas` was found to be overstated by 3.8–8.7 percentage points across all eight states relative to `Birthplace_Elsewhere_P / Tot_P_P` in Table G01, and was corrected before any modelling was run. `pct_bachelor_degree_plus` initially appeared to mismatch when divided by the DataPack's own qualification-table total (`P_Tot_Total` in Table G49B, which excludes respondents with no non-school qualification), but matched exactly once divided by total population aged 15 and over instead, confirming the original figure was correct and identifying the correct denominator for reproducibility. A population_density_km2 indicator, present in an earlier version of this table, was removed: land area is not a Census variable and would have required a separate ABS/Geoscience Australia geographic reference product outside the DataPack release. It was replaced with c6_fortnight_avg (Methods, Policy data).

10.5 Appendix E: Variable Definitions

The following variables are used in this project. Each variable's name, description, and role are defined here.

- face_mask_wearing — **outcome variable** (binary: 1 = compliant, 0 = not). Built by averaging four mask-scenario survey items on a 1–5 scale and thresholding at 4. This is what every model predicts. Also called **mask compliance** in the report.

- survey_week — the number of two-week periods elapsed since the survey began (range 6–51). Captures the overall trend as mandates were progressively introduced.

- age — respondent age in years (mean 48.0, SD 17.1). Older respondents tend to report higher perceived personal risk.

- gender — Male, Female, or Other. Gender differences in compliance are documented in the literature (Haischer et al. 2020).

- household_size — number of people in the respondent's household (1–8). Larger households have more within-home exposure, potentially increasing motivation to protect others.

- employment — five levels: Full time, Part time, Retired, Unemployed, Not working. Employment type determines whether a respondent regularly enters shared spaces. Also called **employment status** in the report.

- wellbeing — a general life-satisfaction score, 0–10. May reflect the degree to which COVID-19 has disrupted daily life.

- perceived_severity — 1–7 scale rating of how serious COVID-19 is perceived to be. Higher perceived severity is expected to increase compliance motivation.

- willingness_to_isolate — 1 (Very unwilling) to 5 (Very willing). A proxy for general willingness to follow public health guidance.

- phq4_status — three levels: Answered, Declined, Not available. Used in place of the underlying mental-health scores, which are unavailable for 23% of respondents (see the PHQ-4 Structural Missingness section in Methods). Also called **mental health response status** in the report.

- `other_behaviour_score` — mean of 13 other protective behaviour items (hand-washing, distancing, etc.) on a 1–5 scale. Captures a general health-behaviour compliance disposition. Also called the **behaviour score** in the report.

- `state` — eight levels, one per Australian state/territory. Used as a plain label in the state-inclusive model to establish the performance ceiling, then replaced by the five measurable predictors below in the measurable-factors model.

- `h6_fortnight_avg` — average government face-covering policy score (0–4) over the respondent's survey fortnight, for their state. Directly measures mask-mandate strictness at that time and place. Also called **mask-policy strictness** in the report.

- `c6_fortnight_avg` — average government stay-at-home requirement score (0–3) over the respondent's survey fortnight, for their state. Measures a second, conceptually distinct policy lever (movement restriction rather than face-covering requirements specifically). Also called **lockdown strictness** in the report.

- `median_weekly_household_income` — 2021 Census state-level median weekly household income (AUD). Captures economic differences between states.

- `pct_bachelor_degree_plus` — 2021 Census percentage of persons aged 15+ holding a Bachelor degree or higher, by state. Higher education is associated with greater public health compliance internationally.

- `pct_born_overseas` — 2021 Census percentage of the state population born overseas. States with higher overseas-born proportions may have had cultural conditions more receptive to mask mandates.